Math 118: Mathematical Methods of Data Theory

Lecture 17: Algorithms for Constrained Optimization

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February 22, 2020

Overview

- Today We'll discuss ways to solve $\min_{\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^d} f(\mathbf{x})$. Specifically:
  - Projected Gradient Descent.
  - Lagrange Methods.
  - Penalty Methods.

Formulating Constrained Optimization Problems

- From Wednesday:

$$\begin{aligned} &\min\ f(\mathbf{x}) \\ \text{subject to: }\ &\mathbf{h}(\mathbf{x}) = \mathbf{0} \\ &\mathbf{g}(\mathbf{x}) \leq \mathbf{0} \end{aligned}$$

- Notationally, convenient to define **feasible region**:

$$\mathcal{X} = \left\{ \mathbf{x} \in \mathbb{R}^d : \mathbf{h}(\mathbf{x}) = \mathbf{0} \text{ and } \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \right\}$$

- Then, original problem becomes $\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$.

Some Examples

- $\min_{\mathbf{x}:A\mathbf{x}=\mathbf{b}} \|\mathbf{x}\|_1$ *(related to sparse recovery/ LASSO)*
- $\min_{\mathbf{x}:\|\mathbf{x}\|_2=1} \mathbf{x}^\top Q\mathbf{x}$ where Q is positive definite.
- $\min_{\mathbf{x}:\mathbf{x}\geq\mathbf{0}} \|A\mathbf{x}-\mathbf{b}\|_2$ *(non-negative least squares)*
- $\min_{\mathbf{x}\in P^d} f(\mathbf{x})$
  - $P^d = \left\{\mathbf{x}\in\mathbb{R}^{d+1} : \mathbf{x}\geq\mathbf{0} \text{ and } \sum_{i=1}^{d+1} x_i = 1\right\}$ is unit simplex.
  - Used when $\mathbf{x}$ denotes a probability measure or in portfolio optimization

Projected Gradient Descent

- Given feasible region $\mathcal{X} \subset \mathbb{R}^d$, define **projection operator:**

$$\Pi_{\mathcal{X}}(\mathbf{x}) = \arg\min_{\mathbf{u} \in \mathcal{X}} \|\mathbf{u} - \mathbf{x}\|_2$$

- Interpretation: $\Pi_{\mathcal{X}}(\mathbf{x})$ returns the point in $\mathcal{X}$ closest to $\mathbf{x}$
- **Projected Gradient Descent:**

$$\boxed{\mathbf{x}_{k+1} = \Pi_{\mathcal{X}}\left(\mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)\right)}$$

- When $\Pi_{\mathcal{X}}$ is exactly computable, Projected Gradient Descent (PGD) has the same convergence rate as GD.
- Questions at this stage:
  - When does $\Pi_{\mathcal{X}}(\mathbf{x})$ exist?
  - When is $\Pi_{\mathcal{X}}(\mathbf{x})$ unique?

The Projection Operator

Theorem

When $\mathcal{X}$ is convex, $\Pi_{\mathcal{X}}(\mathbf{x}) = \arg\min_{\mathbf{u}\in\mathcal{X}} \|\mathbf{u}-\mathbf{x}\|_2$ exists and is unique.

Proof.

- The function $f(\mathbf{u}) = \|\mathbf{u}-\mathbf{x}\|_2$ is strongly convex.
- For strongly convex $f(\mathbf{u})$ and convex $\mathcal{X}$, the problem $\arg\min_{\mathbf{u}\in\mathcal{X}} f(\mathbf{u})$ has a unique solution.
- Thus, $\mathbf{u}^* = \Pi_{\mathcal{X}}(\mathbf{x}) = \min_{\mathbf{u}\in\mathcal{X}} f(\mathbf{u})$ is unique.

□

- **Question:** When is $\Pi_{\mathcal{X}}(\mathbf{x})$ easy to compute?

Examples of Projection Operators

- **Linear Variety:** $\mathcal{X} = \left\{\mathbf{x} \in \mathbb{R}^d : \ A\mathbf{x} = \mathbf{b}\right\}$
  - Exact formula for Π hard to work out.
  - Let $P = I_{d\times d} - A^\top (AA^\top)^{-1} A$.
  - **Claim:** If $\mathbf{x}_k$ satisfies $A\mathbf{x}_k = \mathbf{b}$ then $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k P \nabla f(\mathbf{x}_k)$ also satisfies $A\mathbf{x}_{k+1} = \mathbf{b}$.
  - *Proof:*
    - $A\mathbf{x}_{k+1} = A\left(\mathbf{x}_k - \alpha_k P \nabla f(\mathbf{x}_k)\right) = A\mathbf{x}_k - \alpha_k AP\nabla f(\mathbf{x}_k$
    - Observe that for any $\mathbf{v} \in \mathbb{R}^d$:

$$\begin{aligned} AP\mathbf{v} &= A\left(I_{d\times d} - A^\top (AA^\top)^{-1} A\right)\mathbf{v} \\ &= A\mathbf{v} - AA^\top (AA^\top)^{-1} A\mathbf{v} = A\mathbf{v} - A\mathbf{v} = \mathbf{0} \end{aligned}$$

    - Conclude: $A\mathbf{x}_{k+1} = A\mathbf{x}_k - \mathbf{0} = \mathbf{b}$

- **Unit Sphere:** $\mathcal{X} = \left\{\mathbf{x} \in \mathbb{R}^d : \ \|\mathbf{x}\|_2 = 1\right\}$
  - $\Pi_{\mathcal{X}}(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|_2}\mathbf{x}$
- **Non-negative orthant:** $\mathcal{X} = \left\{\mathbf{x} \in \mathbb{R}^d : \ \mathbf{x} \geq \mathbf{0}\right\}$
  - This is an easy case.
  - $P_{\mathcal{X}}(\mathbf{x})_i = \begin{cases} x_i & \text{if } x_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$
- **Unit Simplex:** $\mathcal{X} = P^d = \left\{\mathbf{x} \in \mathbb{R}^{d+1} : \ \mathbf{x} \geq \mathbf{0} \text{ and } \sum_{i=1}^{d+1} x_i = 1\right\}$.
  - This is a little tricky.
  - See https://arxiv.org/abs/1101.6081 for a good method.

Projected Gradient Method: Conclusions

$$\boxed{\mathbf{x}_{k+1} = \Pi_{\mathcal{X}}\left(\mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)\right)}$$

- For certain well-studied $\mathcal{X}$, $\Pi_{\mathcal{X}}$ is easy to compute.
- In other cases it can be as hard as the original problem.
- When $\Pi_{\mathcal{X}}$ is exactly computable:
  - If $f(\boldsymbol{x})$ convex and L-Lipschitz differentiable:

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_*) \leq \frac{2L\|\boldsymbol{x}_0 - \boldsymbol{x}_*\|^2}{k}$$

  - If $f(\boldsymbol{x})$ μ-strongly convex and L-Lipschitz differentiable we in addition have:

$$\|\boldsymbol{x}_k - \boldsymbol{x}_*\|_2^2 \leq (1 - \alpha\mu)^k \|\boldsymbol{x}_0 - \boldsymbol{x}_*\|_2^2$$

Lagrange Method

- For equality constrained problems:

$$\begin{aligned} &\min f(\mathbf{x}) \\ &\text{subject to: } \mathbf{h}(\mathbf{x}) = \mathbf{0} \end{aligned}$$

- Lagrangian $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\top \boldsymbol{h}(\boldsymbol{x})$
- **Recall:** $\boldsymbol{x}_*$ is sol. to original problem if $(\boldsymbol{x}_*, \boldsymbol{\lambda}_*)$ is stationary point of $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda})$:

$$\nabla_{\boldsymbol{x}} \mathcal{L} = \nabla f(\boldsymbol{x}) + D\boldsymbol{h}(\boldsymbol{x})^\top \boldsymbol{\lambda} = \mathbf{0} \quad \text{and} \quad \nabla_{\boldsymbol{\lambda}} \mathcal{L} = \boldsymbol{h}(\boldsymbol{x}) = \mathbf{0}$$

- **Idea:** Gradient Descent (GD) finds stationary points. Use GD for $\boldsymbol{x}$ variable and Gradient **Ascent** for $\boldsymbol{\lambda}$ variable:

$$\begin{aligned} \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k - \alpha_k \nabla_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}_k, \boldsymbol{\lambda}_k) \\ \boldsymbol{\lambda}_{k+1} &= \boldsymbol{\lambda}_k + \beta_k \nabla_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{x}_k, \boldsymbol{\lambda}_k) \end{aligned}$$

- Can show (c.f. C & Z Chpt. 23) that this converges to $(\boldsymbol{x}_*, \boldsymbol{\lambda}_*)$

Penalty Method

$$\min_{x \in \mathcal{X}} f(x)$$

- Suppose we have function $Q : \mathbb{R}^d \to \mathbb{R}$ with property that:

$$Q(x) = \begin{cases} +\infty & \text{if } x \notin \mathcal{X} \\ 0 & \text{if } x \in \mathcal{X} \end{cases}$$

- Then:

$$x^* = \arg\min_{x \in \mathcal{X}} f(x) \quad \Leftrightarrow \quad x^* = \arg\min f(x) + \gamma Q(x)$$

- **Note:** We have replaced a constrained problem with an unconstrained one.
- **Penalty Function:** Any Q with $Q(x) = \begin{cases} > 0 & \text{if } x \notin \mathcal{X} \\ 0 & \text{if } x \in \mathcal{X} \end{cases}$
- **Penalty Method:** Solve $\min f(x) \quad \Leftrightarrow \quad x^* = \arg\min f(x) + \gamma Q(x)$ for appropriate choice of γ.

Example: Lasso

- In applications, frequently want to solve $A\boldsymbol{x} = \boldsymbol{b}$ where $A \in \mathbb{R}^{m \times n}$ and $m < n$.
- Underdetermined, so need to add additional condition to get unique solution:

$$\min_{\boldsymbol{x}: A\boldsymbol{x} = \boldsymbol{b}} \|\boldsymbol{x}\|_1$$

- Note that $Q(\boldsymbol{x}) = \|A\boldsymbol{x} - \boldsymbol{b}\|_2$ is a penalty function here.
- Apply **Penalty Method**, consider:

$$\min \|\boldsymbol{x}\|_1 + \gamma \|A\boldsymbol{x} - \boldsymbol{b}\|_2 \Leftrightarrow \gamma \left(\min \frac{1}{\gamma} \|\boldsymbol{x}\|_1 + \|A\boldsymbol{x} - \boldsymbol{b}\|_2 \right)$$

$$\Leftrightarrow \min \|A\boldsymbol{x} - \boldsymbol{b}\|_2 + \lambda \|\boldsymbol{x}\|_1$$

- Often, only need $A\boldsymbol{x} \approx \boldsymbol{b}$, penalty method is **flexible enough** to allow this.